

# Can a computer reading a company's annual report tell which ones turned out to be committing accounting fraud?

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*A pre-registered novelty study on SEC 10-K filings.*

**Trained:** five small autoencoders in PyTorch, one per held-out fraud cohort.

**Reused:** a pretrained MiniLM transformer, frozen, as a sentence-to-vector encoder.

**Tested:** six historical accounting frauds; methodology locked in git before any number was looked at.





*Does an unsupervised autoencoder trained on industry-and-year-matched clean 10-K MD&A text assign elevated reconstruction error to known historical fraud filings, under strictly out-of-sample evaluation?*

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The one-line answer this report defends:

At  $N = 6$  frauds, with strict pre-registration: no useful autoencoder signal across four of five evaluable cohorts; Enron is not evaluable under the frozen rule; the one cohort that ranks high (Lehman) is  $n = 1$ , and a 1990s TF-IDF baseline ranks it higher.

# The contract is the contribution.

*git-tagged validation-spec-frozen, committed before any validation script ran.*

## Out-of-sample, always

Leave-one-cohort-out (LOCO): for each held-out fraud, the autoencoder is trained on clean filings from other cohorts only. The fraud and its peers are never in the training set.

## Time-controlled training

Within the LOCO training set, only filings dated on or before that fraud's filing date are eligible. The autoencoder's training corpus cannot include post-fraud disclosure language. (MiniLM pretraining is a separate caveat.)

## Single pass, no tuning

Architecture, hyperparameters, sentence cap, seeds: all committed and tagged before validation ran.  
scripts/o6\_validate.py runs once. No model tuning after validation.

*The contract makes the null result interpretable, not conclusive.*

git tag: **validation-spec-frozen**  
at 98d2585 · 2026-05-01 13:42:42 EDT

## Only Lehman ranks above random expectation; the aggregate still loses to random.

| Cohort                             | Rank    | Note                                                            |
|------------------------------------|---------|-----------------------------------------------------------------|
| Lehman (LEH, FY 2007) <sup>△</sup> | 3 / 13  | <i>n = 1, baseline beats autoencoder</i>                        |
| HealthSouth (HRC)                  | 7 / 12  | at or below random                                              |
| Valeant (VRX)                      | 8 / 13  | at or below random                                              |
| Tyco (TYC)                         | 9 / 13  | at or below random                                              |
| WorldCom (WCOM)                    | 12 / 13 | at or below random                                              |
| Enron (ENE, FY 2000)               | —       | <i>not evaluable: filed earliest, no eligible training data</i> |

### Aggregate hit-rates

hit@1: **0.00** vs random 0.08

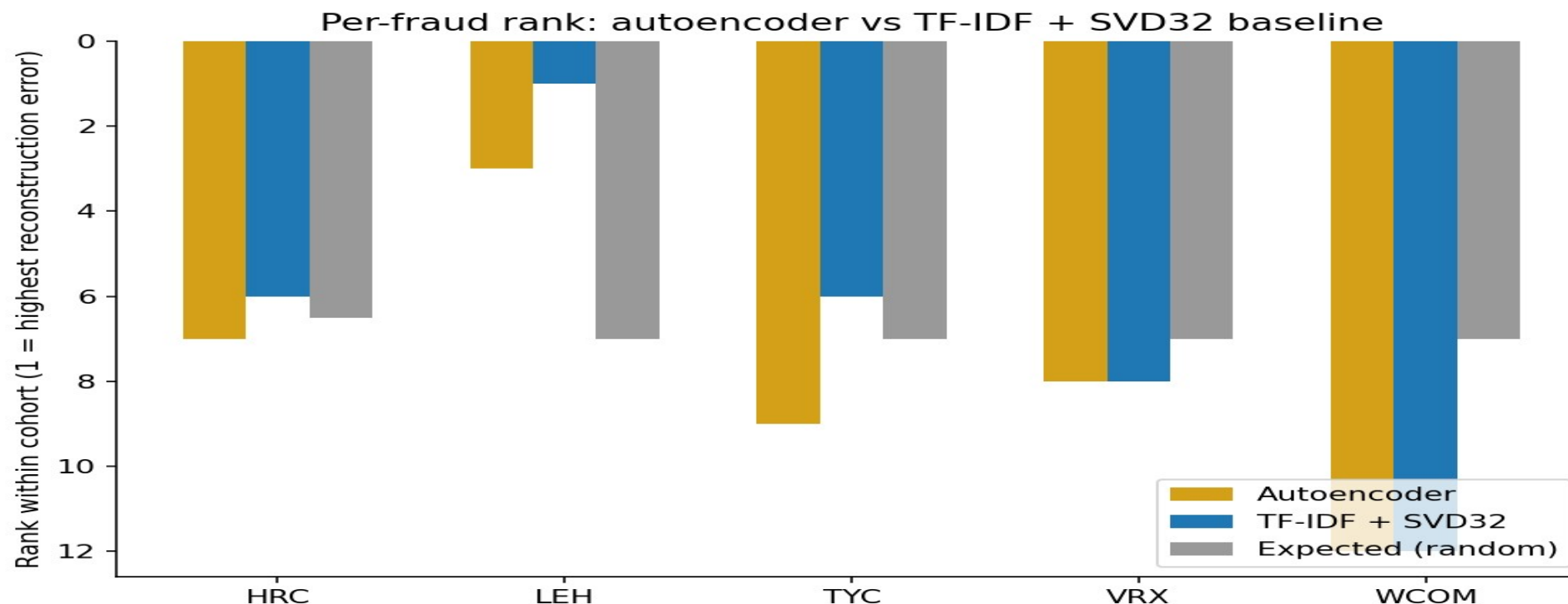
hit@3: **0.20** vs random 0.23

hit@5: **0.20** vs random 0.39

<sup>△</sup> Lehman is the only rank better than random expectation, and a 1990s baseline ranks it higher (slide 5). The aggregate is entirely Lehman.

## A 1990s baseline matched or beat the autoencoder on every cohort.

*Post-hoc: tests whether the autoencoder earned its complexity, does not rescue the primary result.*



*How to read the chart: shorter bar = lower rank number = better detection (rank 1 is most anomalous). Lehman: autoencoder rank 3/13, TF-IDF rank 1/13. Aggregate hit@5 is 0.40 (TF-IDF) vs 0.20 (autoencoder). The pre-registered model does not beat the trivial baseline.*

**Q Did you check a trivial baseline?**

Yes. TF-IDF + truncated-SVD, same bottleneck dimension, same training corpora. It matches or beats the autoencoder on every cohort. The honest claim is that the autoencoder adds nothing here. Slide 5.

**Q The baseline is post-hoc. Isn't that HARKing (hypothesizing after results)?**

The pre-registered hypothesis was about the autoencoder, and the autoencoder result is reported unchanged. The TF-IDF baseline was added after seeing the null and is reported as post-hoc. The primary null does not depend on it.

**Q Could MiniLM have memorized 'Lehman' / 'Repo 105' / etc.?**

Tested post-hoc. Masked 93 such mentions with [ENTITY] and re-scored. Lehman's rank held at 3/13 (delta = 0). Exact-token name leakage did not explain the rank; topical leakage remains open.

## CONCLUSION

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**What survived: pre-registration.**  
**What failed: the autoencoder beyond the TF-IDF baseline.**

*A result that loses to a baseline is not evidence for the neural model. The contract held; the model did not.*

## Next experiment

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Same autoencoder, within-firm temporal delta. Does Lehman's MD&A look anomalous against Lehman's own prior-year MD&A?

Scan to try the pipeline



Pipeline demo (not a fraud scanner): [canary-psi.vercel.app/scan](https://canary-psi.vercel.app/scan)

Code: [github.com/ArseniiChan/canary](https://github.com/ArseniiChan/canary) · Tag: [validation-spec-frozen](#)

*Thanks.*